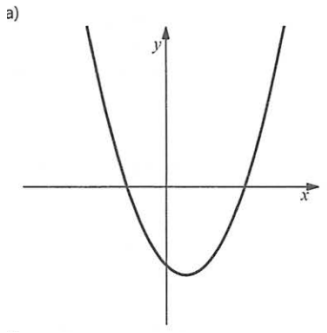
Graph Sketching: Modulus Functions

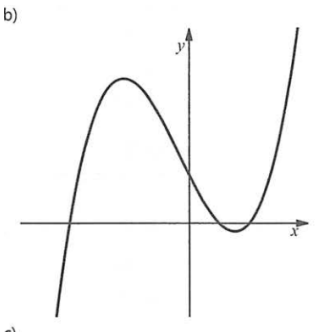
Questions 1–12

Question 1. For each of the following graphs of $y = f(x)$, sketch the graph of $y = |f(x)|$:

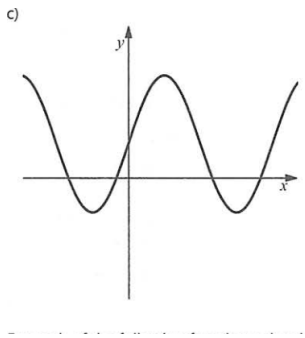
a)



b)



c)

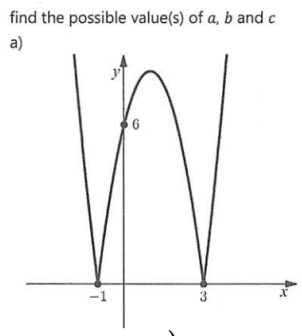


Question 2. For each of the following functions, sketch $y = |f(x)|$:

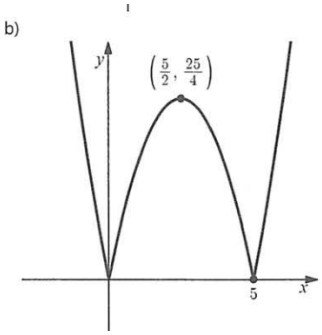
a) $f(x) = x^2 + x - 6$
b) $f(x) = x^2 + 7x + 10$
c) $f(x) = -x^2 + 2x + 8$
d) $f(x) = 6 + 4x - 2x^2$

Question 3. Each of the following graphs are curves of the form $y = |ax^2 + bx + c|$ where a, b and c are integer constants to be found. In each case find the possible value(s) of a, b and c.

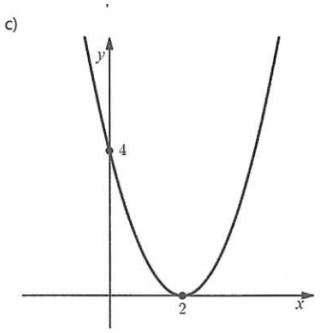
a)



b)



c)

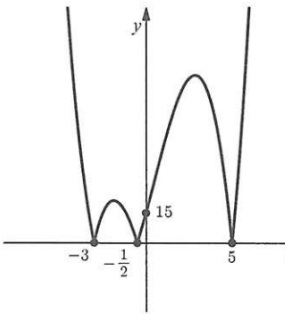


Question 4. In each case, sketch $y = |f(x)|$:

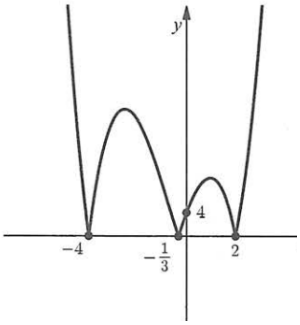
a) $f(x) = x^3 - 9x$
b) $f(x) = x^3 + x^2 - 12x$
c) $f(x) = x^3 - 7x + 6$
d) $f(x) = 14x - 5x^2 - x^3$

Question 5. Each of the following graphs are curves of the form $y = |ax^3 + bx^2 + cx + d|$ where a, b, c and d are constants to be found. In each case find the possible value(s) of a, b, c and d.

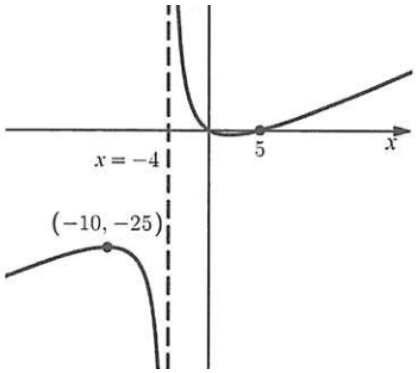
a)



b)



Question 6. The diagram below shows part of the curve with equation $y = f(x)$.



Sketch $y = |f(x)|$.

Question 7. In each case, sketch $y = |f(x)|$:

a) $f(x) = \dfrac{2x+3}{x-1}$

b) $f(x) = \dfrac{1-4x}{2x+5}$

Question 8. In each case, sketch $y = |f(x)|$:

a) $f(x) = \ln(x-1)$

b) $f(x) = \sin x$ for $0 \leq x \leq 2\pi$

c) $f(x) = 1 - \cos x$ for $0 \leq x \leq 2\pi$

d) $f(x) = 1 - e^x$

e) $f(x) = \tan(2x)$ for $0 \leq x \leq \dfrac{\pi}{2}$

Question 9. In each case, by drawing a suitable sketch find the number of solutions for each of the following:

a) $|\ln 2x| = 2x$

b) $\left|\frac{1}{x}\right| - 1 = x^2$

c) $|\sin x| = \frac{2}{\pi}x$

d) $|\cos x| = \frac{2}{\pi}x$

Question 10. In each case, draw a suitable sketch to find the number of solutions for each of the following:

a) $|\sinh x| = 2$

b) $2 - |\cosh x| = 1$

Question 11. Solve each of the following:

a) $|\sinh x| = 1$

b) $|\cosh x - 2| = 1$

Question 12. The curve below has equation $y = \left|\dfrac{ax+b}{cx+d}\right|$ where a, b, c and d are non-zero integers. Determine the values of a, b, c and d giving the values with the smallest possible integer coefficients.

